IC101

Introductory Securities & Markets

Lecture 4 · Money & liquidity

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From last week to this week

We kept saying 'money'. Today: what is it, actually?

Last week we leaned on money

'Deposits ARE money.' 'Banks create money when they lend.'

And a bank run is in most cases really a LIQUIDITY crisis, the cash simply wasn't there.

But we never stopped to ask: what IS money, and what IS liquidity?

I promise to pay the bearer on demand the sum of ten pounds (IOU)



It costs a few pence to make. It is not backed by gold. The government will not swap it for anything.

So why will anyone hand you £10 of real goods for it?

Bottom line: The answer is not gold and not the law. It is something far more fragile and powerful: TRUST. Money works only because we all believe it does (a shared social contract).

Quick poll, grab your phones

What actually makes a £10 note worth £10?

- **A** Gold held in a vault to back it
- **B** The government will swap it for something valuable
- **C** We all simply believe and agree it is worth £10
- **D** The cost of the paper and ink it is printed on

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Today: money, liquidity, and the future

Four parts: what money is, who issues it, how liquid it is, and what comes next

What money is Its three jobs and why it is really just trust, not gold.

The hierarchy of money Types of money and why not all 'money' is equally safe.

Liquidity How fast you can turn something into cash and why that can harm.

The future of money The death of cash, crypto, stablecoins and the digital pound.

What is money? Anything that does three jobs

It is defined by what it DOES, not what it is made of



Medium of exchange

Everyone accepts it, so you can swap it for anything. No need to barter.



Store of value

It holds its worth over time, so you can save today and spend next year.



Unit of account

It is the common ruler we price everything in: a coffee is £3, a car is £30,000 etc.

Do all three jobs and you get money. Strange things have in the past, because a community agreed to trust them:

- **Gold & silver coins**
- **Cigarettes (WWII POW camps)**
- **Cattle**
- **Cowrie shells**

Run the three-job test

Only what passes all three is money

Run the three-job test	Medium of exchange will people take it as payment?	Store of value does it hold its worth?	Unit of account do we price things in it?	Verdict
Cash (£10 note)	✓	✓	✓	**Money** the state's promise
Bank deposit	✓	✓	✓	**Money** 97% of the total
Gold	✗	✓	✗	**Asset, not money** no till takes it
Bitcoin	✗	✗	✗	**Speculative asset** price moves 10% a week
Stablecoin (USDC)	~	~	✗	**Digital dollar IOU** if the reserves are real
Cigarettes, POW camp 1944	✓	~	✓	**Money, there and then** the camp agreed

✓ does the job ~ partly, or only in some places ✗ fails

Do all three jobs and you are money. Nothing is money because of what it is made of.

The money we use: a recap from Lecture 3

Cash is 3% of money. The other 97% is bank deposits, mostly created by lending

Broad money: all the money in the UK, about £3.3 trillion (M4, August 2026)

BANK DEPOSITS, about £3.2tn (97%): IOUs of commercial banks

Notes and coins, £97bn (3%): IOUs of the Bank of England

Narrow money: the part only the Bank of England creates, about £745bn

RESERVES £648bn

Reserves are the accounts commercial banks hold at the Bank of England.
Banks use them to pay each other. You and I cannot hold them.
Both bars are drawn to the same scale.

Notes £97bn + reserves £648bn

Where did the £3.2tn of deposits come from?

Mostly from commercial banks lending: each new loan creates a new deposit (Lecture 3).

Some from the Bank of England's QE: it bought gilts and paid for them with new deposits.

So the money you use every day is mostly a promise from a private bank, not from the state.

Sources: Bank of England, M4 and weekly Bank Return, 2026. Figures rounded.

Your £10 note

an IOU of the CENTRAL bank

"I promise to pay the bearer",
signed for the Bank of England.
A promise from the state. It cannot fail.

Your bank balance

an IOU of a COMMERCIAL bank

A promise from Barclays or Lloyds
to pay you. It spends the same,
but a private bank can fail.

Same spending power, different promisers. That difference is what the rest of today is about.

So what backs the money? Trust

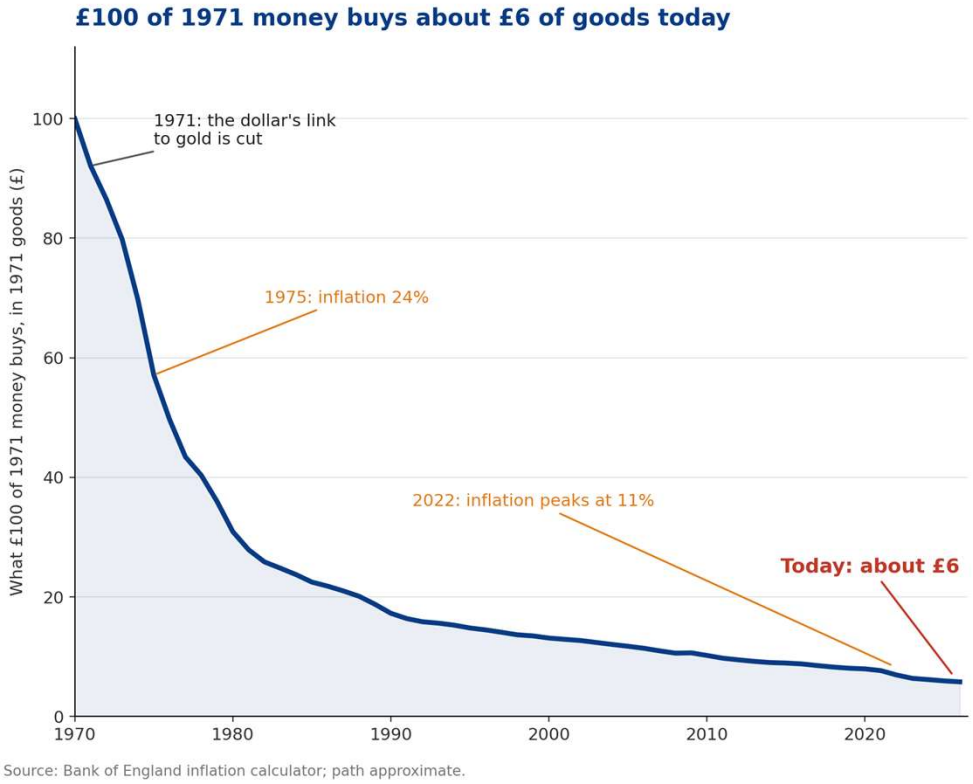
The story of how money cut its final link to gold

Ancient	**Commodity money** Money WAS something valuable: gold and silver coins, worth their weight.
1800s-1971	**The gold standard** Paper notes, but each one swappable for a fixed amount of gold at the bank.
1971	**Nixon cuts the link** The US ends gold convertibility. Since then, money is backed by nothing but trust.
Today	**Fiat money** 'Fiat' = 'let it be'. It is money because the state says so and we all believe it.

Bottom line: When trust dies, money dies: Zimbabwe printed a 100-trillion-dollar note; Venezuela and 1920s Germany the same. Trust is everything.

The quiet tax: inflation

A small dose on purpose, every year. Who pays it, and what happens in overdose



Source: Bank of England inflation calculator; path approximate.

Why it is called a tax

Inflation = prices rising, so each £1 buys a little less every year. The Bank of England targets 2%: money is designed to leak slowly.

Who pays: anyone holding cash or lending at a fixed rate. Your savings account lost buying power every year since 2021.

Who gains: borrowers, and the biggest borrower is the state. Inflation shrinks the national debt in real terms without a vote or a bill. That is why it is quiet.

The spike you lived through: 11.1% in October 2022. £100 in 2020 buys about £80 of goods today.

Same mechanism, bigger dose: more money chasing the same goods. The next slide shows what happens when the dose has no limit.

DEEPER READ UK inflation at 2.9%, far below its 11% peak (ONS, July 2026) ↗

Chart: Bank of England inflation calculator, path approximate. UK CPI peak 11.1% in October 2022; 2.9% in the 12 months to July 2026.

When trust dies: three cases in numbers

Print without limit and the money stops doing all three jobs

Weimar Germany, 1923

3.7 days

how long prices took to double at the peak

Monthly inflation hit about 29,500%.
People burned banknotes for heat;
a loaf of bread cost a wheelbarrow of cash.

Why it printed

to pay war reparations and striking workers

Print without limit and the trust goes.

Zimbabwe, 2008

79,600,000,000%

inflation in one month, November 2008

Prices doubled every 25 hours.
The state printed a 100-trillion-dollar note,
then gave up its currency for the US dollar.

Why it printed

to cover spending after farm output collapsed

Print without limit and the trust goes.

Venezuela, 2018

about 1,000,000%

inflation in one year (estimates vary)

The bolivar became near-worthless.
Shops weighed banknotes instead of
counting them; wages were paid daily.

Why it printed

to fund deficits as oil income fell

Print without limit and the trust goes.

1 Foreign income collapses
oil, farms or reparations: no dollars, so no imports, and the shelves empty

→

2 Only one tap is left
tax revenue falls with the economy and nobody will lend, so wages are paid by printing

→

3 Same wages, fewer goods
prices rise, so next month's wage bill is bigger, so the state prints more

→

4 Everyone dumps the currency
people spend it the day they get it or buy dollars; money stops doing all three jobs

Bottom line: Nobody chooses hyperinflation. A state with no exports, no tax base and no lenders chooses to pay this month's wages by printing, and chooses it every month. The cure is a central bank that can say no, which is why the Bank of England is independent.

The family tree of money

From gold coins to the numbers on your phone

Commodity money

Valuable in itself: gold & silver coins ('specie').

Representative

Paper claims on a commodity. The old gold-backed note.

Fiat money

Backed only by trust and the state. Today's cash and coins.

Electronic / bank money

Digital deposits, ~97% of it. Numbers on a bank's computer.

Near-money & e-money

Easily turned to cash: PayPal, M-Pesa, wallets, new crypto claimants.

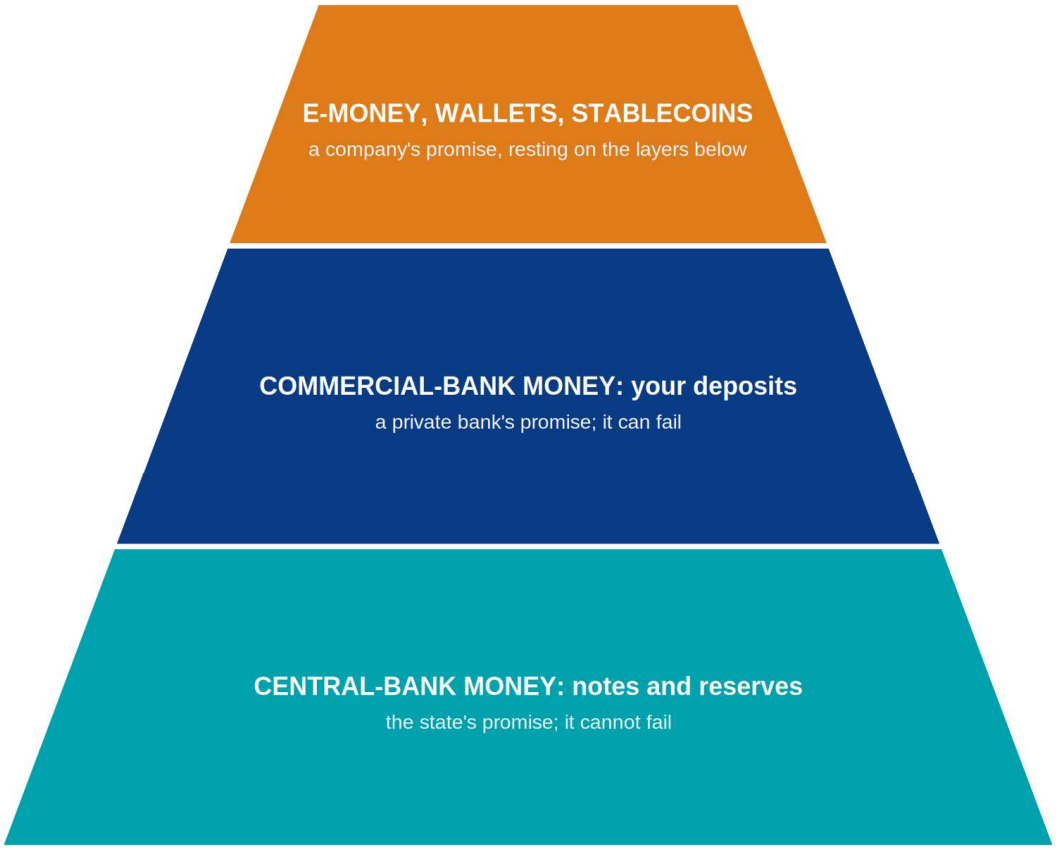
The trend is clear:

money keeps getting more abstract, from gold you can hold, to a claim, to trust, to a number, to a line of code.

CLICK **Explore the types of money with AI** ↗

The pyramid, with real UK numbers

The safest layer is the smallest: about £745bn of state money under £3.2tn of bank money



SAFER AS YOU GO DOWN.

A bank run is everyone rushing down the pyramid into cash at once.

How big each layer really is (UK, 2026)

£ billions, drawn to scale

Notes in circulation

£97bn

Banks' reserves at the Bank of England

£648bn

Bank deposits (broad money less cash)

£3,200bn

E-money and stablecoins held in the UK

small: est. tens of £bn

The pyramid is upside down in size.

The safest money, the state's, is the smallest layer: about £745bn. The layer we all use, deposits, is four times bigger. That gap is why a run is dangerous: not everyone can get down to the base.

Sources: Bank of England weekly report, March 2026; M4, August 2026; FCA estimate.

Liquidity: how fast can you turn it into cash?

The second great idea of today, and the quiet killer of banks

The definition, and the price of it

Liquidity = how quickly and cheaply you can turn an asset into cash without dropping its price.
Cash is perfectly liquid: it IS cash. A house is illiquid: to sell it fast you must slash the price.

Why it matters: bills come due in cash, not in assets. A family, a firm or a bank can own plenty and still fail to pay on Friday, because what it owns cannot be turned into cash in time. Every bank in Lecture 3 that failed was in that position, long before it ran out of value.

Why it matters to you: it is the hidden question behind every investment. Not only "what will it be worth?" but "can I get out when I need to and at what price?"

Two things it is NOT

Not the same as wealth. You can be rich but illiquid, all your money tied up in property.

Not the same as solvency. A bank can be solvent (sound) yet illiquid (no cash today), exactly the killer distinction from Lecture 3.

Bottom line: **Liquidity is like air: you never think about it, until suddenly there is none, and then it is the only thing that matters!**

Quick poll, grab your phones

Which of these could you turn into cash the FASTEST at full value?

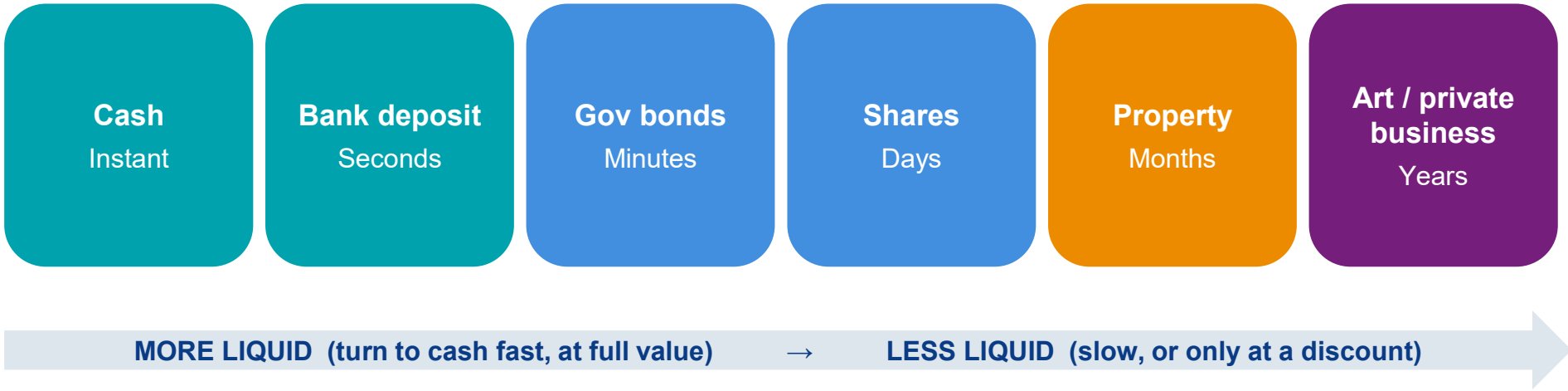
- **A** A £250,000 flat you own
- **B** £5,000 of shares in a big listed company
- **C** £5,000 in your current account
- **D** A £5,000 painting



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The liquidity spectrum

Everything you own sits somewhere on this line



The price of liquidity. Less-liquid assets pay more: the liquidity premium. A 10-year gilt yields less than a loan to a small firm partly because you can sell the gilt in a second. The premium is your reward for tying money up, and your compensation for the risk that you cannot get out when you need to. In a panic that premium jumps: in March 2020 even the gap between buying and selling a US Treasury bond widened tenfold.

Why it matters beyond banks. A bank borrows at the liquid end (instant deposits) and lends at the illiquid end (25-year mortgages). Funds do the same: a property fund promising daily withdrawals owns buildings that take months to sell, which is why UK property funds had to shut their doors to withdrawals in 2016 and again in 2020. Liquidity risk is the gap between what you have promised and what you actually hold.

When the liquidity vanishes

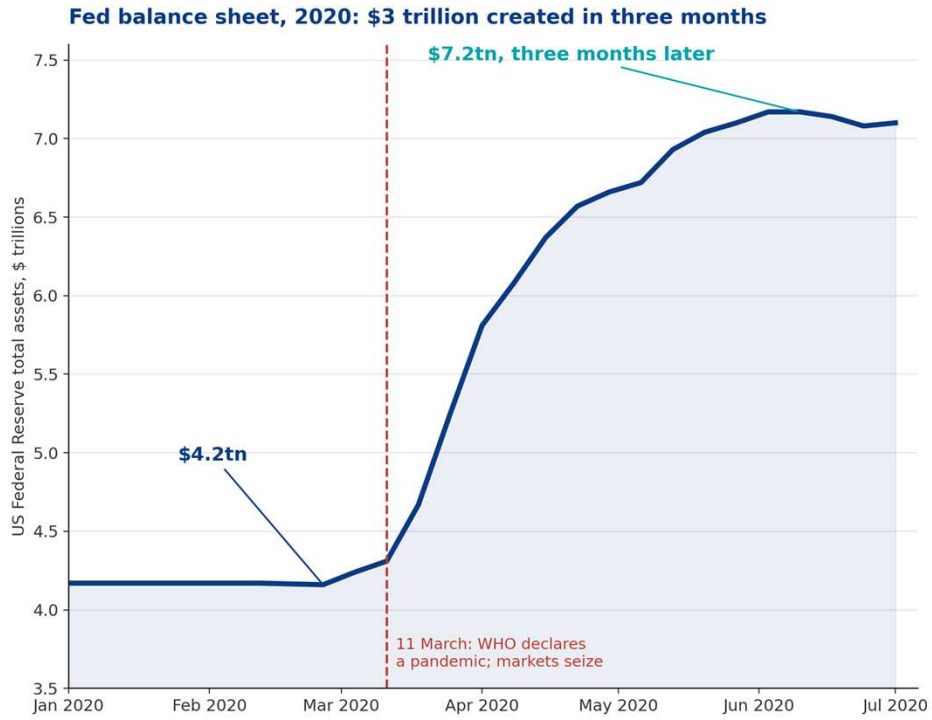
It is always there, until the one moment you need it

Three kinds of run, one mechanism

A bank run: the bank is solvent, its loans are fine, but it cannot turn them into cash fast enough. The central bank, lender of last resort, lends cash against those good loans.

A market run (right): March 2020, everyone sold at once (liquidated), even US Treasuries. Only a central bank can make more cash, so the Fed bought $3tn of bonds in three months and the market reopened.

A fund run: September 2022, UK pension funds had to sell gilts to meet collateral calls; gilt prices fell, forcing more selling. The Bank of England bought £19bn of gilts in two weeks to stop the spiral.



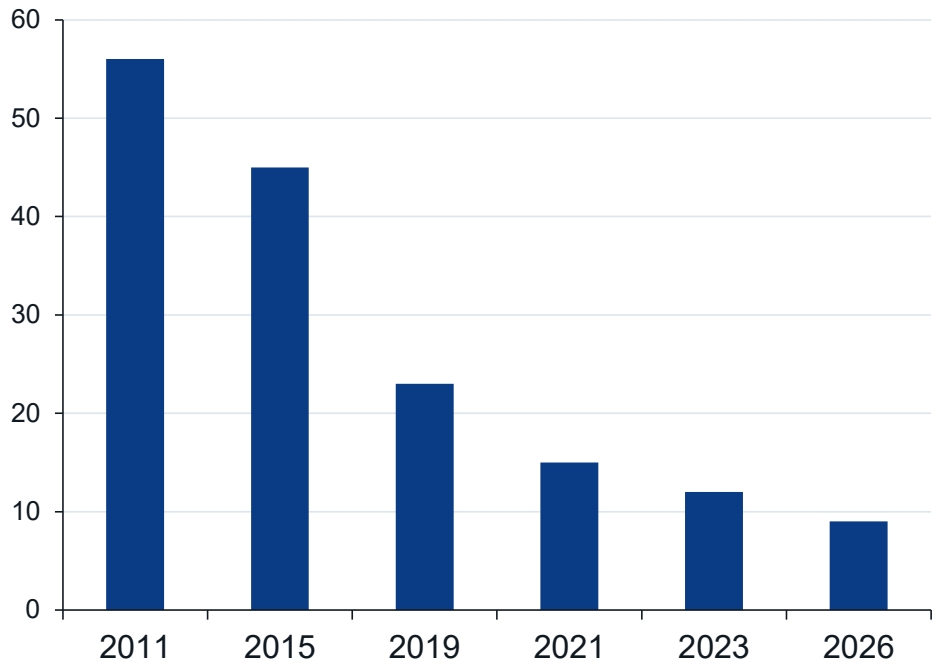
Source: Federal Reserve H.4.1, weekly, approximate. Bank of England: £450bn of QE announced in 2020.

Bottom line: Liquidity is confidence. The moment everyone doubts it and reaches for cash together, it evaporates, for banks, funds and whole markets alike.

The slow death of cash

In a decade, Britain went from a cash society to a tap society

Cash as a share of UK payments %



What is replacing it

Cash fell from over half of UK payments in 2011 to below 10% today. Cards overtook cash back in 2017.

Over half of UK adults now use a mobile wallet (Apple Pay, Google Pay).

Sweden is nearly cashless. Kenya's M-Pesa put a bank in every phone.

But cash still matters: for the vulnerable, for privacy, and when the power or network fails.

DEEPER READ Cash falls below 10% of UK payments for the first time (UK Finance) ↗

UK cash share of payments: UK Finance, UK Payment Markets. Mobile wallet adoption: UK Finance, 2024-25.

What happens when you tap?

The plumbing behind a cashless payment

You tap and it just works. Underneath, four steps move the money:

1 You tap
card or phone at the till

2 Authorise
the card network (Visa/Mastercard) or Faster Payments checks the money is there, in seconds

3 Deposits move
your bank debits your account; the shop's bank credits theirs

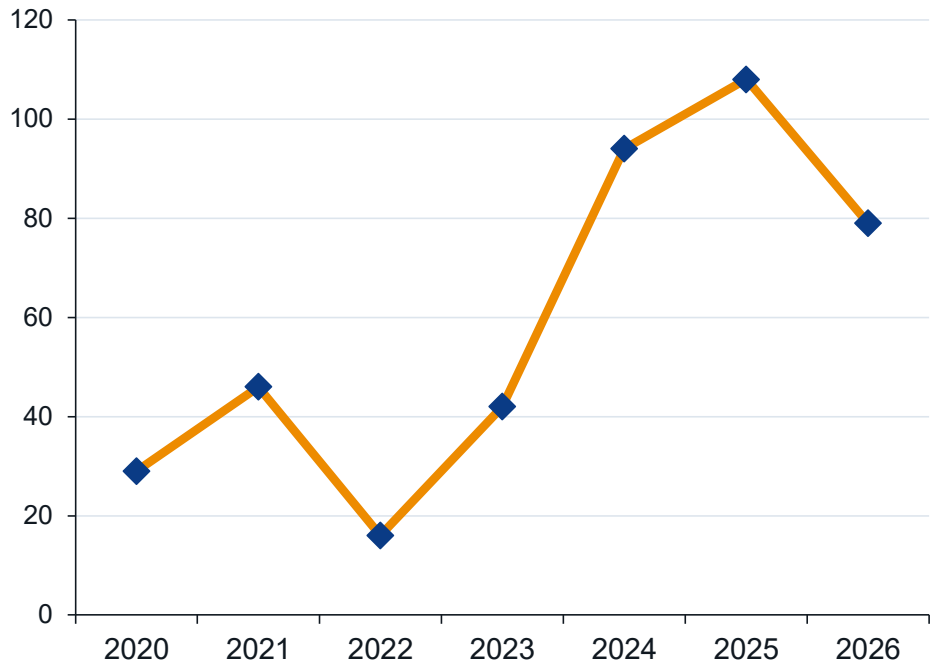
4 Settle
the two banks settle the net difference in Bank of England reserves; the Bank's settlement system moves about £350bn a day

Bottom line: Nothing physical moves. A payment just reassigns bank deposits, and the banks settle in central-bank money, the base of the pyramid. That is why the base still matters as cash disappears.

Challenger 1: crypto and Bitcoin

Born in 2009 out of the bank bailouts: money with no bank, no state and no printing

Bitcoin price $ thousands



Why it exists, and is it money?

The idea: launched January 2009, weeks after the bailouts. No bank, no state, no central bank that can print: a fixed supply of 21 million coins, run by code and a crowd.

Run the test. Medium of exchange: few shops take it. Store of value: about $126k in October 2025, about $79k by September 2026. Unit of account: nobody prices a coffee in it.

Verdict: today it fails all three. It behaves like a speculative asset, "digital gold", not money.

Real-world test: El Salvador made it legal tender in 2021; people took the free $30 and went back to dollars; the law was dropped in 2025 at the IMF's request.

DEEPER READ Bitcoin selloff leaves half of all supply trading at a loss (Bloomberg, June 2026) ↗

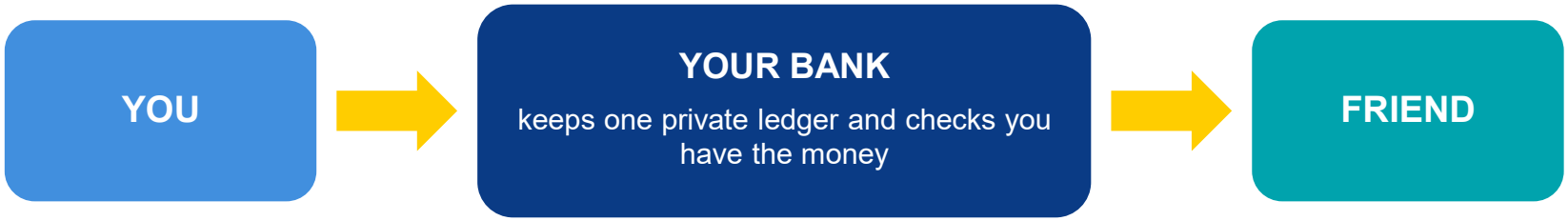
CLICK Is Bitcoin money? Ask AI ↗

Bitcoin approx prices; all-time high ~$126k in Oct 2025, ~$79k by September 2026 (about 37% below peak). Invented 2008 by 'Satoshi Nakamoto'.

How does Bitcoin actually work?

Take out the bank in the middle; put in a shared public record everyone can see

The normal way: a trusted middleman keeps the record



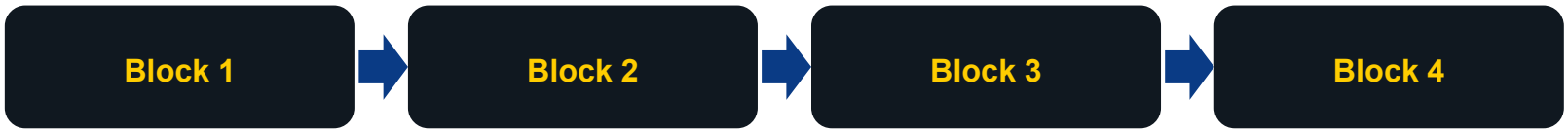
You must trust the bank to hold the record honestly.

The Bitcoin way: no middleman, everyone shares one public record



No bank. You trust open code and the crowd, not an institution.

The 'blockchain' = verified payments sealed into blocks, each locked to the one before, so nobody can rewrite history:



Challenger 2: stablecoins

Crypto's answer to its own volatility problem

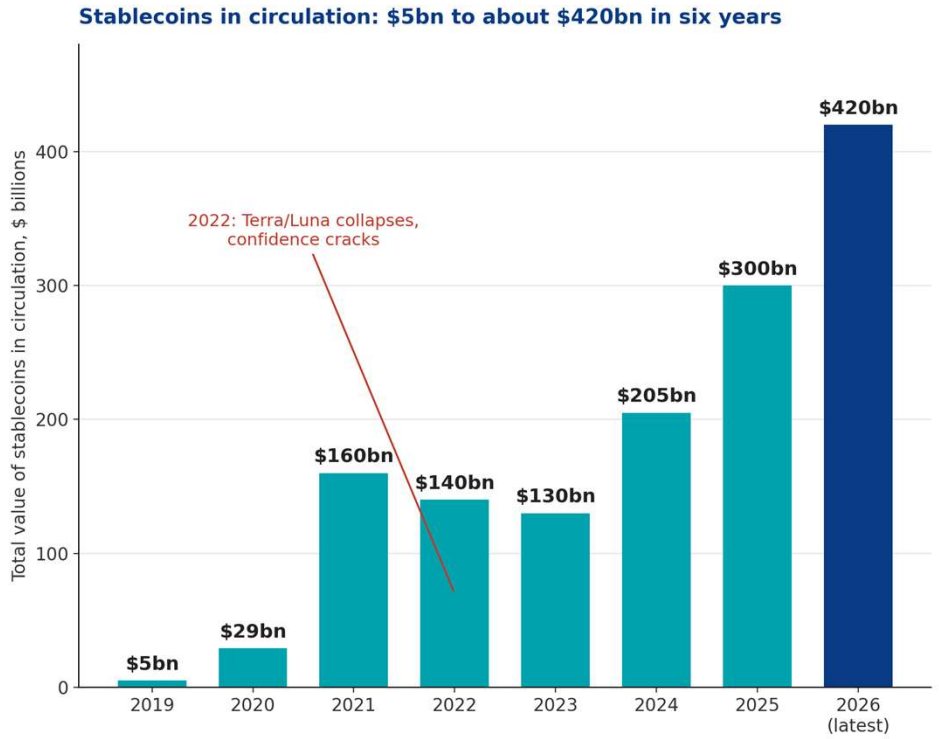
What it is, who uses it, who gains

What: a crypto token pegged 1-to-1 to the dollar. $1 in, one token out, and the issuer holds the dollar in reserve, in Treasury bills and bank deposits.

Who uses it: crypto traders (the cash leg of most Bitcoin trades); savers in Argentina, Turkey and Nigeria holding digital dollars their banks cannot give them; migrants sending money home for cents instead of 6%; firms paying freelancers abroad.

Who issues it and why: Tether and Circle (USDC). They keep the interest on the reserves, billions a year, and pay holders nothing. The US Treasury gains a new buyer of its debt, which is why Washington legalised them in 2025.

The catch: no deposit insurance. Terra, with no real reserves, went to zero in May 2022; USDC fell to 87 cents when its bank, SVB, failed in 2023.



Source: DefiLlama / CoinGecko, approximate year-end totals.

DEEPER READ Wall Street pushes back on the trillion-dollar stablecoin boom (Bloomberg, July 2026) ↗

CLICK How do stablecoins hold their peg? Ask AI ↗

Challenger 3: the digital pound (a CBDC)

The central bank's own digital cash, for everyone

What a CBDC is

A **Central Bank Digital Currency**: digital money issued by the central bank itself, that YOU could hold directly.

Today only banks hold central-bank money. A CBDC opens the base of the pyramid to everyone.

~130 countries (98% of world GDP) are exploring one. China's e-CNY is live; the UK is designing a 'digital pound'.

Why bother, if nobody is asking for one?

1 A public option. When the last cash machine closes, every pound you hold is a private bank's promise. A CBDC keeps state money available to the public: digital cash.

2 Who controls payments. The trigger was Facebook's Libra in 2019. The ECB wants out of dependence on Visa, Mastercard and dollar stablecoins; China wanted the state back between Alipay, WeChat and every payment.

3 Cheaper plumbing. Card fees cost UK shops over £1bn a year; a CBDC could be near-free, instant and work offline when networks fail.

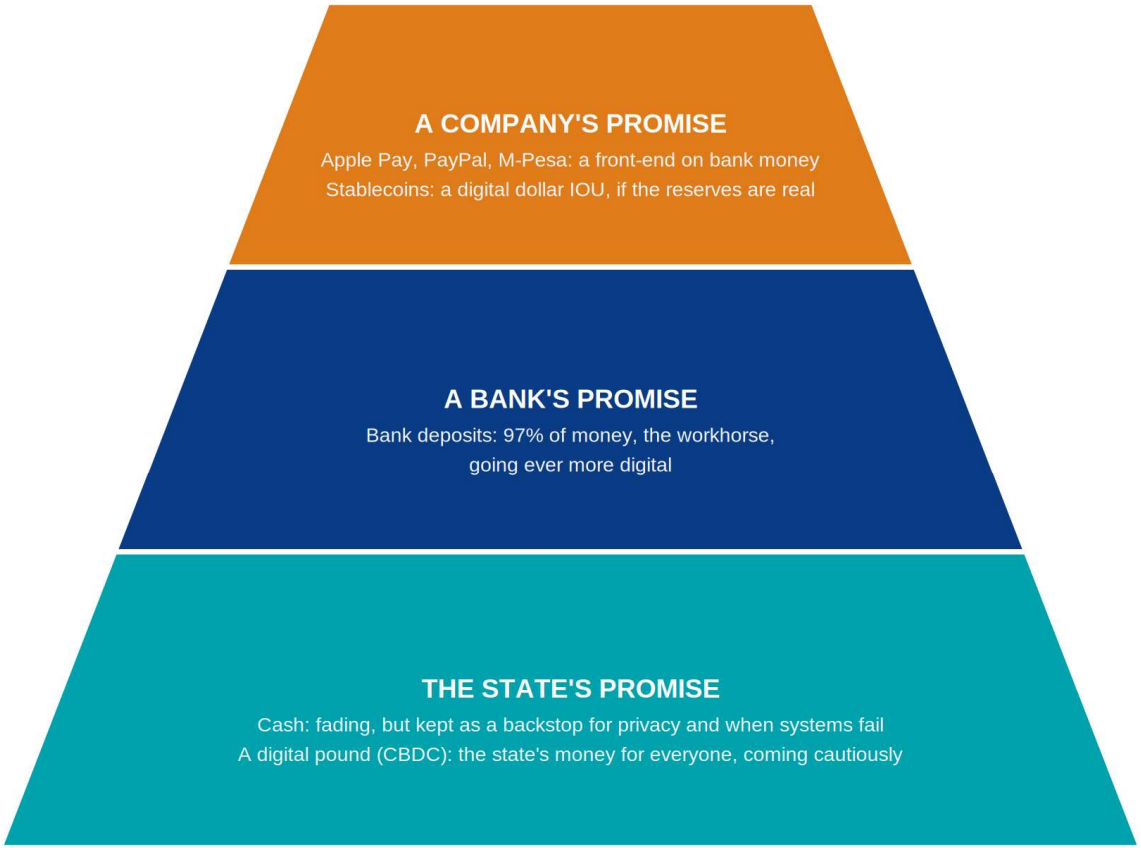
The peril: privacy (the state could see every payment) and the banks (deposits could drain into it, making a run instant). So the design caps holdings at £10,000 to £20,000 and pays no interest: a backstop, not a savings account.

DEEPER READ Digital pound update: the Bank's 2026 design phase and decision timeline (Bank of England) ↗

CLICK The digital pound: explore the debate with AI ↗

So what wins? Ask whose promise it is

Place each contender by who stands behind it. Bitcoin stands outside the pyramid



Whose promise is it? That tells you how safe it is.

The winner will be whatever we trust most.

NOBODY'S PROMISE

Bitcoin and crypto

No issuer, no state, no reserves: just code and a crowd.
Fails the three-job test. A speculative asset, not money.

It sits outside the pyramid entirely.

The contest, as it stands

Cash	under 10% of UK payments and falling
Deposits	97% of money; not going anywhere
Wallets	over half of UK adults use one
Stablecoins	about $420bn, the fastest growing
Bitcoin	volatile; dropped as legal tender by El Salvador
CBDC	130+ countries exploring; China live, UK designing

Quick poll, grab your phones

A UK 'digital pound' launches. Your biggest concern would be…

- **A** Privacy, the state could see my spending
- **B** It could destabilise the banks (easier runs)
- **C** Nothing, I would happily use it
- **D** I'd stick with cash and normal accounts

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The highlights, know these cold

1. Money is defined by three jobs: medium of exchange, store of value, unit of account.
2. Money is not currency: cash is ~3%, the other ~97% is digital bank deposits.
3. **Modern money is fiat, backed by nothing but trust. Inflation is a quiet tax on holding it; hyperinflation is that tax without a brake.**
4. The money pyramid: central-bank money (safest) > commercial-bank money > e-money.
5. **Liquidity = how fast & cheaply you turn an asset into cash; illiquid assets pay a liquidity premium; it is not wealth, not solvency.**
6. A bank run is a liquidity crisis: everyone rushing DOWN the pyramid into cash at once.
7. Cash is dying (from over half of UK payments in 2011 to under 10% now); the future is a contest of digital money.
8. Bitcoin fails as money (too volatile); stablecoins peg to $; CBDCs are state digital cash.

Your mid-term multiple-choice test is built from this slide.

Exam check, can you answer these?

Same style as your mid-term test

Which is NOT one of the three jobs of money?

A) medium of exchange B) store of value C) making a profit

Answer: C. The three jobs are medium of exchange, store of value, unit of account.

An asset is 'liquid' if it…

A) is worth a lot B) can be turned into cash fast, at full value C) pays high interest

Answer: B. Liquidity is about speed and cost of turning it into cash, not its value.

Bitcoin is best described today as…

A) a stable global currency B) central-bank money C) a volatile, speculative asset

Answer: C. It fails the money test on volatility; it behaves like 'digital gold'.

Where we are, and where we're going

Lecture 4 of 5

1	The big picture & the three pillars	✓ done
2	Balance sheets & financial intermediaries	✓ done
3	Commercial, investment & central banks	✓ done
4	Money & liquidity	✓ today
5	Capital flows & the four sectors	→ next week

You now understand money and liquidity, the lifeblood. **Next, the finale: how that money flows between households, firms, governments and the world.**

Over to you

Before next week

- Read Chapter 5 (pp. 79-91).
- Look ahead: Chapter 6 (sectors).
- Read the BIS 'Future of Money' article (Blackboard).
- Take the Blackboard quiz.

The one idea to keep

Money is trust. A £10 note, a bank balance, gold, even Bitcoin: all are just different things we agree to trust.

Liquidity is how fast you can turn something into cash, and losing it is what brings banks down.

YOUR SEMINAR THIS WEEK **Seminar 4: Money**

It goes deeper on today: what money is, fiat and specie, e-money and near-money, and central bank digital currencies.

To prepare: read the BIS 'Future of Money & Payments' article and Chapter 5.